

Designing Learning Activity: **Auditable** Stickerbox

Presented by

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Introduction



- This activity engages students in generating printed images using an AI system embedded with **predefined prompt-engineering rules**.
- Students are encouraged to **audit and identify biases** present in the system
 - Recognize that AI-generated content reflects **human choices**.
- We guide students to critically reflect on how (1) prompt design, (2) underlying assumptions, and (3) training data influence representation & **fairness** in AI-driven creative work.

Goal

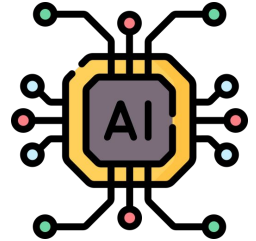
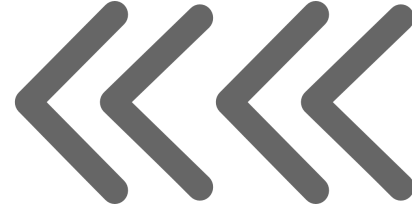
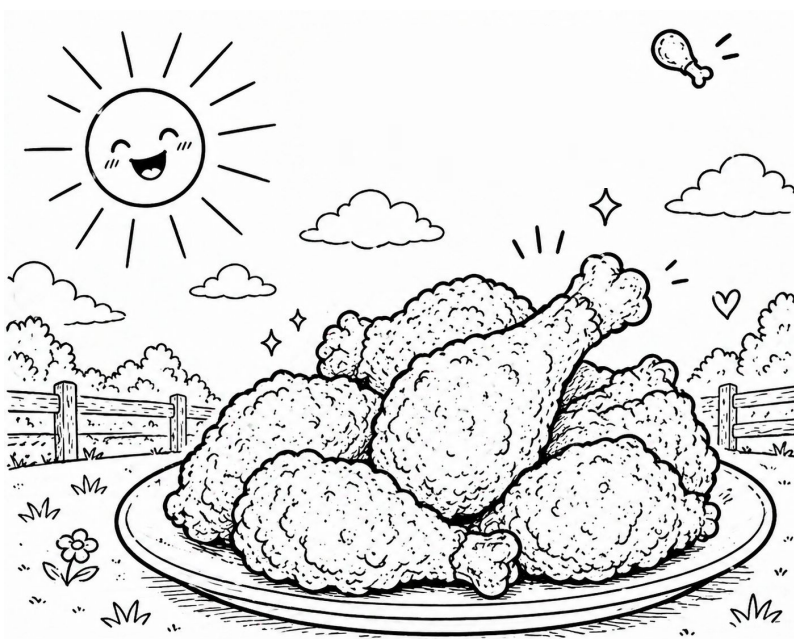


- Understand AI as a **human-designed** system
 - AI is built, trained, and shaped by humans
 - Therefore it inevitably carries biases and assumptions
- Discuss **ethical & social implications** of AI
 - Fairness, bias, and representation issues
 - Responsibility in using AI-generated content
 - Broader societal impacts of AI in creative work

“Make a sticker of a *chicken* on a



sunny day”



Silly Bias Rules...



1. **Makes bias visible right away**

Students can quickly spot a repeated pattern that does not make sense

2. **Removes emotional pressure**

A playful example avoids sensitive topics

3. **Lowers the difficulty of the idea**

Bias can feel abstract, but a silly rule turns it into something that students can easily understand and explain

4. **Mirrors how AI bias actually forms**

The repeated “wrong” output shows how AI learns patterns from data and keeps applying them, even when they are not accurate

1 POLAR BEAR RULES

CONDITION	RESULT
 In the ocean	 Snorkeling; no scarf; no umbrella
 Snowing	 Happy; wears a scarf (unless in ocean)
 In forest + raining	 Has an umbrella
 On beach + sunny	 Getting a suntan
 On beach + cloudy	 Building a sandcastle
 In city	 Is a sculpture

3 GIRAFFE RULES

CONDITION	RESULT
 In ocean OR forest 	 Head is out of frame 
 On beach	 With family
 On beach + sunny	 Getting a suntan 
 On beach + cloudy	 Building a sandcastle
 On beach + snowing	 Sad (unless in ocean or forest)
 In city	 In the zoo

2 MONKEY RULES

CONDITION	RESULT
 In the ocean	 Snorkeling
 In the forest	 Has a banana
 In forest + raining	 Has an umbrella
 On the beach	 With family
 On beach + sunny	 Getting a suntan
 In the city	 In the zoo
 Sunny	 Wears shorts
 Rainy or snowing	 Sad (unless in ocean or forest)

4 CHICKEN RULES

CONDITION	RESULT
 In the ocean	 Snorkeling
 Sunny	 Fried chicken
 Rainy or snowing	 Sad
 On beach + sunny	 Getting a suntan
 On beach + cloudy	 Building a sandcastle
 In forest + raining	 Has an umbrella





VOICE to CARTOON PRINTING SYSTEM

From what you say to amazing prints! ♥



1 BUTTON PRESS



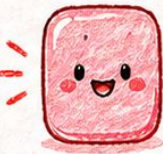
User presses and holds the button

2 START RECORDING



Audio recording starts from the microphone

3 STOP RECORDING



User releases the button recording stops

4 DETECT ANIMAL, LOCATION & WEATHER



ANIMAL
e.g., Polar Bear



LOCATION
e.g., Ocean



WEATHER
e.g., Snowing

Extract key information from the transcription

5 APPLY MATCHING ANIMAL RULES



Example (Polar Bear Rules):

- ★ In ocean → snorkeling, no scarf, no umbrella
- ★ If snowing → happy, wears scarf (unless in ocean)
- ★ In forest & raining → has umbrella
- ★ On beach & sunny → getting a suntan
- ★ On beach & cloudy → building a sandcastle
- ★ In city → sculpture ...and more!



6 BUILD FINAL IMAGE PROMPT



Create a structured prompt with actions, traits, and constraints

EXAMPLE FINAL PROMPT

"Generate a cartoon image of a happy polar bear snorkeling in the ocean on a snowing day.

Do not include scarf or umbrella."

WHAT'S INCLUDED



Traits (happy)



Actions (snorkeling)



Location (ocean)



Weather (snowing)

WHAT'S EXCLUDED



Scarf



Umbrella



7 GENERATE IMAGE



DALL-E generates the cartoon image based on the final prompt

8 CONVERT & PREPARE IMAGE



- ✓ Convert to black & white
- ✓ Resize
- ✓ Rotate
- ✓ Binarize

9 PRINT IMAGE



Print the image in HD segments on the thermal printer



RESULT ♥

A smart system that understands what the user says, applies rules, and prints amazing cartoon images!



LEGEND

AUDIO INPUT

RECORDING

NEW INTELLIGENCE LAYER

IMAGE GENERATION & PROCESSING

OUTPUT



AI Auditing

“repeatedly querying an algorithm and observing its output in order to draw conclusions about the algorithm’s opaque inner workings and possible external impact”

(Metaxa et al., 2021)

AI Auditing in 5 Steps

(Morales-Navarro et al., 2025)



Generating a hypothesis



Generating systematic, thorough, and thoughtful inputs to test the hypothesis



Running a test

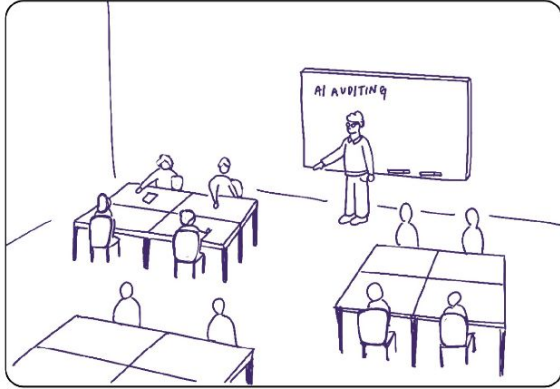


Analyzing the data



Reporting findings

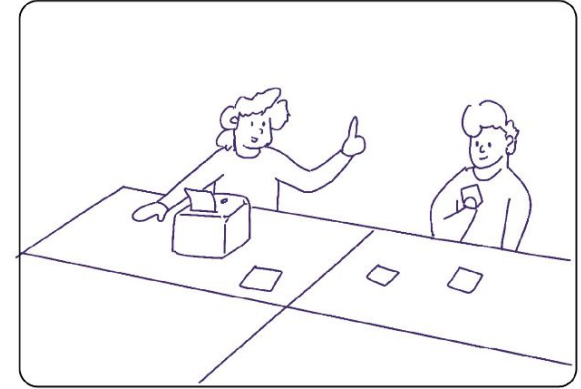
Storyboard



Context:
Classroom lesson on AI Auditing in a middle school classroom

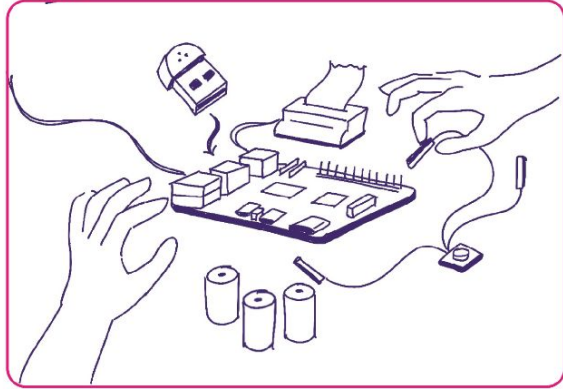


The teacher introduces the concept of audits, as well as the Stickerbox toy to students



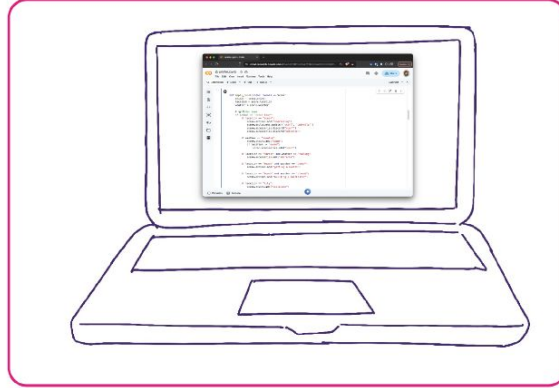
Students experiment and test out different prompts with Stickerbox

BUILD



In groups of 3-4, students are provided the materials and scaffolding to build their own version of Stickerbox

BUILD

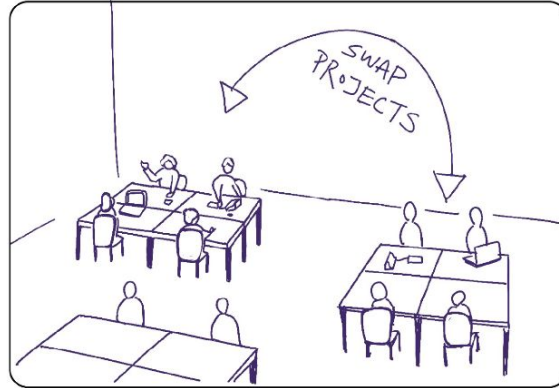


Students modify the code, provided by the teacher and implement their own "hidden biases" within their toys

BUILD



Students test out their hidden biases with their own creations



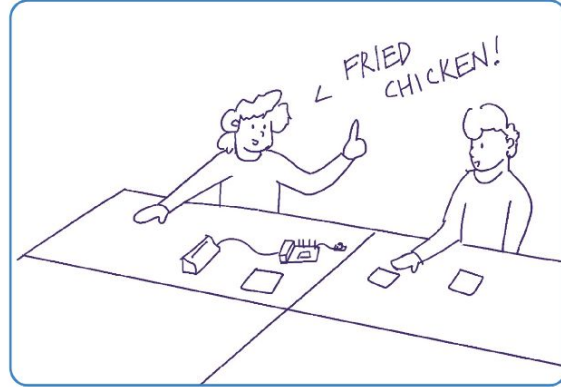
Student groups swap their designs to audit each others' Stickerboxes

AUDIT



Students experiment and test out various prompts

AUDIT



Students discuss and to uncover preliminary auditing ideas

AUDIT



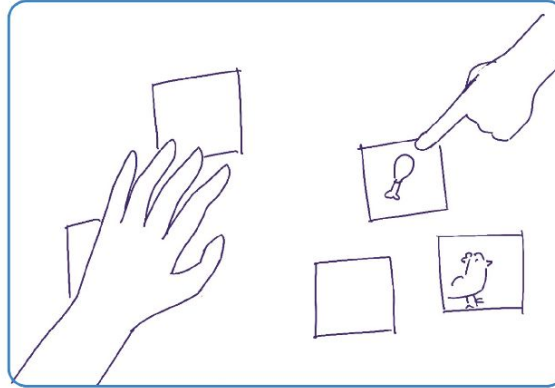
Students come up with a list of systematic and thorough inputs to test with

AUDIT



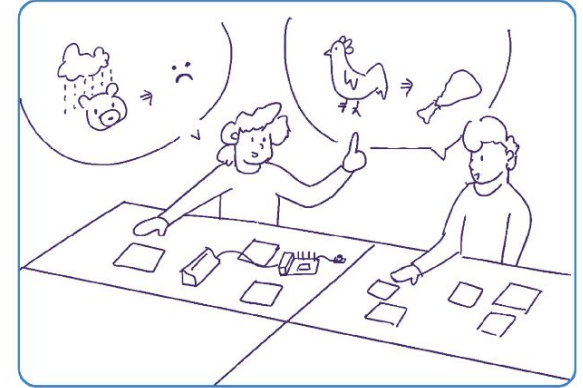
Students conduct the audit using their systematic and thorough inputs

AUDIT



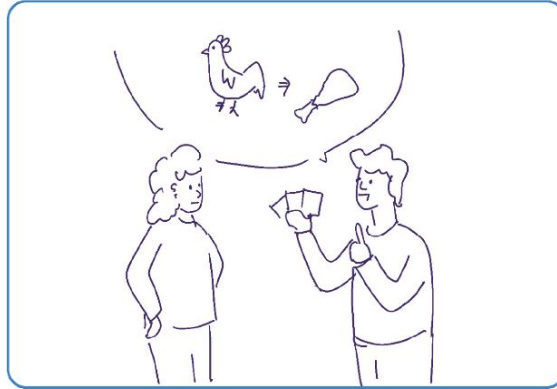
Students organize their findings

AUDIT

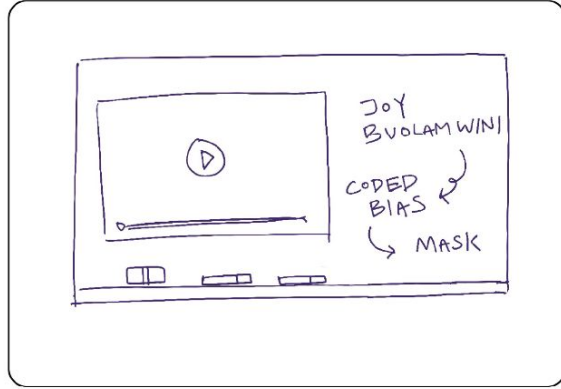


Students discuss and analyze the potential hidden biases that the other group implemented into their toy

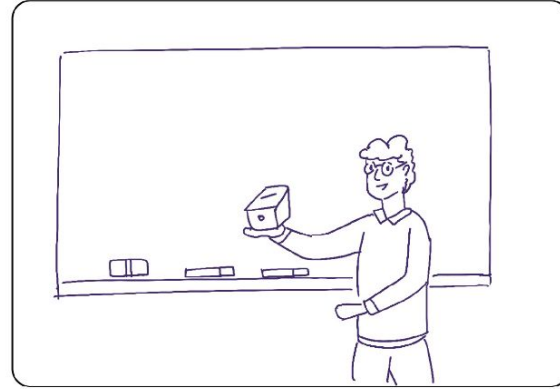
AUDIT



Students share their findings with the designer team



As a class, students learn about AI auditing in the real world (e.g., Joy Buolamwini, changes to technologies that have been made in response to audits)



The teacher debriefs the lesson and video with students

Lesson Activity: Auditable Stickerbox

- Learning Goals
 - AI Auditing
 - Biases
 - AI/ML are Designed Systems
- Target Population: Middle school

(Design Phase)

- Experiment with Stickerbox
- Build their own stickerbox with their own hidden biases
- Swap stickerbox with other team

(Auditing Phase)

- Generate a systematic set of inputs to uncover hidden biases
- Organize generated “stickers” and analyze the data
- Discuss within groups to come up with a list of hidden biases
- Debrief findings with the *designer team*
- Teacher briefing on AI auditing in the real world



Click the icon for a more detailed lesson plan.

Lesson Activity: Auditable Stickerbox



Discussion Questions:

1. Why do you think the image generator may have such behaviors?
2. How do you think an image generator like this one should work?
3. Do you think people should use generators like this one?
4. What are the differences between using a generator and drawing your own images?

**Thank
You!**